

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

First Semester (2026-2027)

Tutorial Test 1 (Closed Book)

CHE F313 (Separation Processes II)

Date: 07/09/2026

Max Time: 15 mins

Max Marks: 15

Name:

ID No:

Sec No:

Q1 320 kW of power is required to crush 140 tonnes/h of a material. If 80% of the feed passes through a 50-mm screen and 80% of the product passes through a 2-mm screen, calculate the work index of the material. What will be the power required for the same feed at 140 tonnes/h to be crushed to a product such that 80% is to pass through a 1.6-mm screen?

Solution:

Bond's equation:-

$$E(\text{kWh/tonne}) = 0.3162 W_i \left(\frac{1}{\sqrt{P_{80}}} - \frac{1}{\sqrt{F_{80}}} \right) \rightarrow [1]$$

case 1:-

$$\text{throughput} = 140 \text{ t/hr}$$

$$\text{power} = 320 \text{ kW}$$

$$F_{80} = 50 \text{ mm}$$

$$P_{80} = 2 \text{ mm}$$

$$E = \frac{320}{140} = 2.2857 \text{ kWh/t} \rightarrow [2]$$

$$\text{size term} = \left[\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{50}} \right] = [0.7071 - 0.1414] = 0.5657 \rightarrow [3]$$

$$\begin{aligned} \therefore \text{work index} = W_i &= \frac{E}{(0.3162 \times 0.5657)} \\ \Rightarrow W_i &= \frac{2.2857}{0.1788} = \boxed{12.78 \text{ kWh/tonne}} \end{aligned}$$

new power for $P_{30} = 1.6 \text{ mm}$

~~new size term~~

$$= \left[\frac{1}{\sqrt{1.6}} - \frac{1}{\sqrt{50}} \right] = [0.7905 - 0.1414] \rightarrow [2]$$
$$= 0.6491$$

specific energy

$$E_2 = 0.3162 W_c (0.6491)$$

$$\Rightarrow E_2 = 0.3162 (12.78) (0.6491)$$

$$\Rightarrow E_2 = 2.623 \text{ kWh/t}$$

$$\therefore \text{power } P_2 = E_2 (140)$$

$$= 2.623 (140) \text{ kW}$$

$$\Rightarrow P_2 = 367.22 \text{ kW}$$

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